MEGATHON X: SINGULARITY · PROBLEM STATEMENT 1 (LLOYDS)

GreenLens AI

An Explainable Green Customer Insights Platform Grounded in Evidence

Team 43 | Retail Customer & Business Intelligence Dashboards

✓ Analyzes everyday transaction patterns to identify four green product opportunities.

✓ Strictly separates machine learning predictions from deterministic carbon calculations and bank policy.

✓ Guarantees zero hallucinations through a constrained, verified explanation architecture.

Why Green Banking Needs Strict Separation

- **The Banking Challenge:** Banks hold rich debit spend and account data, but lack tools to pinpoint genuine green transition opportunities.
- **The Greenwashing Risk:** Unverified carbon numbers destroy regulatory and consumer trust. Impact must be auditable and conservative.
- **Dual Stakeholder Value:** Retail customers get non-intrusive green guidance; relationship managers get high-priority portfolio leads.
- **Our Standard:** Practical insights that never confuse commercial sales propensity with credit eligibility or financial approval.

Retail Customer

Clear, personalized green options (EV, Solar, Green Home, Paperless) with full reason codes and customer memory controls.

Relationship Team

Portfolio intelligence across 5,000 customers with validated priority screens and impact drill-downs.

Compliance & Governance

Audit-ready operational impact scenarios with documented UK government emission factors and zero LLM guesswork.

The Four-Way Division of Responsibility

- **Machine Learning Predicts:** Estimates simulated customer take-up propensity from spending trends and circumstances.
- **Carbon Engine Calculates:** Computes operational emission impacts using verified 2025 UK government factors.
- **Product Rules Decide Eligibility:** Enforces explicit bank policies (e.g. existing EV exclusions) independently of ML scores.
- **The LLM Only Explains:** Acts as a safe sentence selector from approved templates; never predicts, calculates, or approves.

1. ML Models
→ Predicts simulated take-up propensity

2. Carbon Engine
→ Computes documented annual kgCO2e

3. Product Rules
→ Decides eligibility & explicit exclusions

4. Guarded LLM
→ Only selects approved factual sentences

A Modular, Read-Only Production Pipeline

- **Immutable Snapshot:** Relational customer data and transactions import into SQLite with SHA-256 source verification.
- **Batch Inference:** Four trained scikit-learn pipelines load once at startup and score 5,000 customers in seconds.
- **Deterministic Carbon Engine:** Verified UK conversion factors compute operational scenarios without database mutation.
- **Guarded Delivery:** Typed FastAPI contracts serve twin React dashboards with 100% offline deterministic fallbacks.

END-TO-END PIPELINE

1. Synthetic CSVs
   5,000 customers, 368k transactions
2. Customer 360
   28 pre-cutoff behavioral features
3. ML Inference
   4 propensity models + TF-IDF classifier
4. Rules & Carbon
   2025 DESNZ factors + exclusion screens
5. Guarded LLM
   Safe sentence selection (fails closed)
6. FastAPI & React
   Customer Insights + Business Intelligence

Synthetic UK Cohort with Clean Splits

- **Synthetic UK Cohort:** 5,000 customers and 368,323 transactions with realistic UK spending distributions (no real PII).
- **Disjoint Partitioning:** 3,500 train (70%) / 750 validation (15%) / 750 test (15%), split strictly by customer ID with 0% overlap.
- **Pre-Cutoff Features:** All 28 Customer 360 features derive strictly from activity prior to the synthetic prediction cutoff date.
- **Zero Leakage:** Numeric missing-value imputers use training medians only, verified by automated audit scripts.

DATASET SPECIFICATIONS

Total Cohort Size:
5,000 synthetic UK customers

Transaction Records:
368,323 category-mapped rows

Training Partition:
3,500 customers (70%)

Validation Partition:
750 customers (15%)

Held-Out Test Partition:
750 customers (15%)

Customer Overlap:
0% (completely disjoint IDs)

Feature Leakage:
Zero (all pre-cutoff features)

Propensity Results & The Unseen-Merchant Check

- **Four Models Beat Baselines (Test AUC):** GP01 EV 0.751, GP02 Home 0.815, GP03 Solar 0.901, GP04 Paperless 0.866.
- **Priority Thresholds (Test F1):** EV 0.389 (thr 0.24), Home 0.327 (thr 0.10), Solar 0.353 (thr 0.23), Paperless 0.577 (thr 0.28).
- **Transaction Categorization:** TF-IDF text model achieves 0.980 test macro F1 on customer splits vs 0.918 for rule baseline.
- **Honest Unseen-Merchant Check:** On 7 held-out merchants, TF-IDF drops to 0.095 while the rule baseline retains 0.921.

PROPENSITY TEST EVALUATION

GP01 EV Finance: AUC 0.751 | F1: 0.389
GP02 Green Home Loan: AUC 0.815 | F1: 0.327
GP03 Solar Finance: AUC 0.901 | F1: 0.353
GP04 Paperless: AUC 0.866 | F1: 0.577

HONEST UNSEEN-MERCHANT DIAGNOSTIC

Customer Split: TF-IDF F1 = 0.980 | Rule = 0.918

Merchant Holdout: TF-IDF F1 = 0.095 | Rule = 0.921

Finding: Classifier memorized merchant names. Production requires keyword/regex rules for new merchants.

Deterministic Scenarios & Honest Boundary Limits

- **Verified UK Factors:** Official 2025 DESNZ factors (petrol 2.069 kg/L, diesel 2.571 kg/L, UK electricity 0.177 kg/kWh).
- **Operational Scenarios:** EV models petrol emissions vs grid charging; Green Home models a 10% gas efficiency package.
- **Solar Regional Boundary:** Models a 4 kWp London reference system (919 kWh/kWp/year yield); all other regions return UNAVAILABLE.
- **Paperless Boundary:** Shows 48 pages/year avoided; CO2 is marked UNAVAILABLE because no verified lifecycle envelope factor exists.

WORKED EXAMPLE: CUSTOMER C002056

Customer Circumstances:
Petrol car, 24,033 annual miles, £289/mo fuel

Baseline Transport Emissions:
5,618 kgCO2e / year (petrol combustion)

EV Scenario Charging Emissions:
1,234 kgCO2e / year (UK grid electricity)

Net Annual Avoided Impact:
4,384 kgCO2e / year (ESTIMATED)

TRANSPARENT BOUNDARIES

- Solar outside London: UNAVAILABLE (no local yield data)
- Paperless CO2: UNAVAILABLE (no verified envelope factor)

Transparent Evidence, Reason Codes & Exclusions

- **Auditable Reason Codes:** Every opportunity surfaces plain reasons (e.g. HIGH_FUEL_SPEND, EXISTING_PETROL_VEHICLE).

- **Evidence Drawer:** Customers and advisors inspect exact inputs (fuel spend £288.70, mileage 24,033) and conversion source URLs.

- **Qualitative Reliability Tiers:** Tiers (High / Medium / Low / Unavailable) assess feature completeness and training familiarity.

- **Dedicated Exclusion Section:** Ineligible customers appear in a separate 'Not recommended' section with clear reasons.

LIVE APPLICATION DEMONSTRATION

Interactive Customer & Business Workspaces

- **Step 1:** Retail Customer View (C002056): Shows Customer 360 profile, EV Finance card (58% propensity), and savings.

- **Step 2:** Eligibility Separation (C000042): High EV propensity is suppressed to EXCLUDED by rule EXISTING_EV_NO_REPLACEMENT.

- **Step 3:** What-If Scenario Explorer (C000018): Toggle EV or Paperless chips to re-run models dynamically without altering records.

- **Step 4:** Business Dashboard: Portfolio view across 5,000 customers with drill-down from segment to evidence and actions.

Guarded LLM Explanations & Ethics by Design

- **Zero Free-Form Hallucinations:** LLM is a constrained style selector choosing between pre-approved factual sentences.
- **Validated Guardrails:** 28/28 offline test cases pass (including 4/4 missing/conflicting data checks); fails closed to templates.
- **Privacy Protection:** Customer IDs and personal markers are stripped before sending any external API payload.
- **Customer Autonomy:** Interactive memory controls allow users to snooze ('Remind me later'), explore, or permanently dismiss.

GUARDRAIL & GOVERNANCE SCORECARD

✓ **Groundedness Guardrails:**
28 / 28 cases passed

✓ **Factual Accuracy Guardrails:**
28 / 28 cases passed

✓ **Completeness Checks:**
28 / 28 cases passed

✓ **Numerical Consistency:**
28 / 28 cases passed

✓ **Missing / Conflicting Data:**
4 / 4 edge checks passed

✓ **Automated Test Suite:**
57 automated tests passed (51 backend + 6 frontend)

Honest Limitations & Next Steps

- **Synthetic Data Boundary:** Metrics reflect the synthetic generator; testing against real banking data is essential before deployment.
- **Merchant Generalization:** Unseen merchants require regex dictionaries and fuzzy category lookups alongside ML models.
- **Solar Geographic Scope:** Solar models London only; expanding nationwide requires localized irradiance and EPC roof data.
- **Future Roadmap:** Integrate PCAF-aligned financed emissions reporting and test live LLM provider responses under banking network SLAs.

CURRENT LIMITS & FUTURE ROADMAP

Current Limit: Synthetic Labels

Performance reflects simulation; real cohort testing is required.

Current Limit: Merchant Holdout

Model memorized synthetic merchant tokens; needs regex fallback.

Next Step: EPC & Solar Ingestion

Ingest UK Energy Performance Certificates for roof orientation.

Next Step: PCAF Accounting

Implement financed emissions attribution metrics for banking.

CONCLUSION

GreenLens AI: Actionable, Transparent Green Banking

- **Meets the Brief:** Full four-product platform with dual customer and relationship manager dashboards.
- **Principled Separation:** Machine learning predicts take-up, deterministic rules compute carbon, and AI explains facts.
- **Audit-Ready Governance:** Honest about limits, zero greenwashing, and fully reproducible across 57 automated tests.
- **Thank You:** Team 43 is ready to take your questions.

KEY TAKEAWAY

GreenLens AI proves that banks can deliver intelligent, personalized green recommendations while upholding the highest standards of financial governance.

- 4 Green Products Implemented
- 5,000 Customers Analyzed
- 100% Deterministic Carbon Engine
- Zero LLM Hallucinations
- 57 Automated Tests Passing

Ready for Questions & Demonstration